Thermal-Aware Power Budgeting on a Silicon Die: A Unified Mathematical Pipeline from Field Conduction to Linear Programming

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***Abstract*—Managing heat in modern multicore chips requires connecting fundamental physics with mathematical optimization. In this work, we present an end-to-end six-stage pipeline that travels from heat-transfer physics to a convex linear program solved in Python. Starting from Fourier's law and energy conservation, we derive the steady-state Poisson equation and apply temporal Laplace transforms to extract dynamic thermal impedance and its steady-state limit. We discretize the continuous field over a 40×40 grid using a five-point finite-difference stencil, assembling a sparse CSR matrix system. An epigraph reformulation transforms the non-smooth min-max peak-temperature objective into a linear program subject to a strict 5.0 W chip budget. When an unmanaged thread contention concentrates 3.2 W on a single core, localized peak temperature reaches 343.2 K. Our linear program redistributes power into a balanced 1.25 W per-core allocation, suppressing the hotspot down to 318.4 K. This achieves a 24.8 K temperature reduction, beating the benchmark target requirement of at least 5.0 K in under two seconds of runtime.**

***Index Terms*—Silicon thermal management, Poisson equation, Laplace transform, finite difference method, sparse linear algebra, linear programming, HiGHS.**

I. Introduction and Physical Scenario

As microprocessors pack more transistors into compact packages, power dissipation creates severe localized heat spots. High temperatures degrade silicon reliability, speed up chip aging, and trigger thermal throttling that degrades performance. Because heat spreads conductively through bulk silicon, a core's temperature depends not only on its own power draw, but also on heat flowing in from adjacent blocks and out through boundary heat sinks.

This capstone project solves the power-allocation problem for a square silicon die with side length $L = 10\,\mathrm{mm}$ and thermal conductivity $k = 150\,\mathrm{W/(m\cdot K)}$. All four die boundaries are held at ambient temperature $T_{\mathrm{amb}} = 300\,\mathrm{K}$ by an idealized heat sink frame. Four active cores ($N_c = 4$) are placed symmetrically at grid coordinates $(10, 10)$, $(10, 29)$, $(29, 10)$, and $(29, 29)$ on a 40×40 grid, corresponding to spatial positions $(2.56\,\mathrm{mm}, 2.56\,\mathrm{mm})$, $(2.56\,\mathrm{mm}, 7.44\,\mathrm{mm})$, $(7.44\,\mathrm{mm}, 2.56\,\mathrm{mm})$, and $(7.44\,\mathrm{mm}, 7.44\,\mathrm{mm})$. The total power dissipated by all cores cannot exceed $P_{\max} = 5.0\,\mathrm{W}$.

Rather than guessing power limits by trial and error, our pipeline traverses six stages across engineering mathematics:

1) **Stage 1 (Vector Calculus):** Derive local heat conduction from Fourier's law and energy balance.
2) **Stage 2 (PDEs):** Formulate and classify the steady-state Poisson boundary-value problem.
3) **Stage 3 (Transforms):** Apply the temporal Laplace transform to define thermal impedance $Z_{\mathrm{th}}(s)$ and its steady-state resistance matrix.
4) **Stage 4 (Linear Algebra):** Discretize the domain using a five-point central difference stencil into a sparse CSR linear system.
5) **Stage 5 (Optimization):** Cast peak-temperature minimization into an epigraph linear program.
6) **Stage 6 (Python Verification):** Solve the LP with HiGHS and verify constraints and heatmaps programmatically.

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II. Mathematical Formulation: Field Model and Poisson PDE

A. Energy Conservation and Fourier's Law

Let $T(\mathbf{x}, t)$ be the local temperature in kelvin, and $\mathbf{q}(\mathbf{x}, t)$ be the conductive heat flux vector in $\mathrm{W/m^2}$. For isotropic silicon with uniform thermal conductivity k, Fourier's law states:

$$\mathbf{q} = -k\nabla T, \tag{1}$$

where ∇T is the spatial temperature gradient. For an arbitrary control volume V, the net conductive heat escaping through boundary ∂V must equal the rate of internal volumetric heat generation Q in $\mathrm{W/m^3}$:

$$\oint_{\partial V} \mathbf{q}\cdot\mathbf{n}\,dS = \int_V Q\,dV. \tag{2}$$

Applying Gauss's divergence theorem transforms the surface flux into a volume integral:

$$\int_V (\nabla\cdot\mathbf{q} - Q)\,dV = 0 \implies \nabla\cdot\mathbf{q} = Q. \tag{3}$$

Substituting (1) into the conservation equation gives:

$$\nabla\cdot(-k\nabla T) = Q \implies \nabla^2 T = -\frac{Q}{k}. \tag{4}$$

Equation (4) completes Stage 1 using vector divergence and gradient.

B. Boundary Conditions and PDE Classification

Assuming a planar steady-state profile across the die domain $\Omega = (0, L) \times (0, L)$, the governing equation becomes:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = -\frac{Q(x,y)}{k}, \quad (x,y) \in \Omega. \tag{5}$$

The outer die edges are kept at constant ambient temperature by an external thermal frame:

$$T(x,y) = T_{\text{amb}} = 300\,\text{K}, \quad \forall\,(x,y) \in \partial\Omega. \tag{6}$$

Evaluating the discriminant $D = b^2 - 4ac$ for general second-order PDE operators $aT_{xx} + bT_{xy} + cT_{yy}$ yields $a = 1, b = 0, c = 1$, so $D = -4 < 0$. This classifies the system as strictly elliptic, which guarantees smooth spatial diffusion. Defining excess temperature rise $\theta(x,y) = T(x,y) - T_{\text{amb}}$ converts the boundary condition into homogeneous Dirichlet data ($\theta = 0$ on $\partial\Omega$). By the elliptic maximum principle, the highest temperature rise must occur strictly inside the die domain where power is injected. This completes Stage 2.

III. Transform-Domain Analysis and Thermal Impedance

Accounting for material thermal capacitance yields the dynamic heat diffusion equation:

$$\rho c_p \frac{\partial T}{\partial t} = k\nabla^2 T + Q(x,y,t), \tag{7}$$

where ρ is silicon density in kg/m^3 and c_p is specific heat capacity in $\text{J/(kg}\cdot\text{K)}$. Taking the one-sided Laplace transform with respect to time t for zero initial excess temperature ($\theta(x,y,0) = 0$) gives:

$$\rho c_p s\Theta(s) = k\nabla^2\Theta(s) + \mathcal{Q}(s), \tag{8}$$

where s is complex frequency in s^{-1}, and $\Theta(s) = \mathcal{L}\{\theta(t)\}$. Rearranging terms highlights the balance between transient energy storage and spatial conduction:

$$(\rho c_p s - k\nabla^2)\Theta(s) = \mathcal{Q}(s). \tag{9}$$

We define the dynamic thermal impedance operator $Z_{\text{th}}(s)$ as the transfer ratio relating temperature rise to injected input power:

$$Z_{\text{th}}(s) = \frac{\Delta\Theta(s)}{P(s)}. \tag{10}$$

Applying the final value theorem as $s \to 0$ drops the dynamic accumulation term $\rho c_p s \to 0$, reducing (9) back to the elliptic Poisson equation:

$$\lim_{s\to 0} Z_{\text{th}}(s) = R_{\text{th}}, \tag{11}$$

where R_{th} represents the steady-state thermal resistance matrix in K/W. This analytical bridge completes Stage 3.

IV. Spatial Discretization and Sparse Matrix Assembly

To solve the Poisson equation numerically, we discretize the $10\,\text{mm} \times 10\,\text{mm}$ die using a uniform $N_{\text{grid}} \times N_{\text{grid}} = 40 \times 40$ grid ($N_{\text{grid}} = 40$ points per axis). The spatial step size is:

$$h = \frac{L}{N_{\text{grid}} - 1} = \frac{0.01\,\text{m}}{39} \approx 0.2564\,\text{mm}. \tag{12}$$

The total grid contains 1600 nodes, with $(N_{\text{grid}} - 2)^2 = 1444$ active interior nodes. Applying a central five-point difference stencil to approximate the Laplacian at an interior node (i,j) gives:

$$\nabla^2 T_{i,j} \approx \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j}}{h^2}. \tag{13}$$

Applying this stencil across all interior nodes yields the sparse linear system:

$$A_{\text{int}}\boldsymbol{\theta} = -\frac{1}{k}B\mathbf{p}, \tag{14}$$

where A_{int} is the 1444×1444 discrete Laplacian matrix, $\boldsymbol{\theta}$ is the vector of interior temperature rises, $\mathbf{p} = (p_1, p_2, p_3, p_4)^\top$ represents the four core powers, and B maps core powers to spatial grid cells.

Because each stencil node connects only to its four direct neighbors, A_{int} has at most five non-zero values per row. We store A_{int} using the Compressed Sparse Row (CSR) format, keeping memory usage minimal. Boundary rows adjacent to the edges incorporate the Dirichlet $300\,\text{K}$ values directly. The discrete Dirichlet energy ensures negative definiteness:

$$\lambda_{\max}(A_{\text{int}}) < 0, \tag{15}$$

which guarantees that the linear system is nonsingular, numerically stable, and has a unique solution. This completes Stage 4.

V. Epigraph Linear Programming Formulation

Our design goal is to minimize the highest temperature anywhere on the die while staying within the total power ceiling $P_{\max} = 5.0\,\text{W}$. Mathematically, this peak objective is non-smooth:

$$\min_{\mathbf{p}} \max_{i} T_i. \tag{16}$$

To make this problem solvable with standard linear solvers, we use an epigraph reformulation. We introduce a scalar helper variable t representing the upper bound on all node temperatures. This turns the non-smooth min-max problem into a clean, smooth linear program:

$$\min_{\mathbf{p},\boldsymbol{\theta},t} \quad t \tag{17}$$

$$\text{subject to} \quad A_{\text{int}}\boldsymbol{\theta} + \frac{1}{k}B\mathbf{p} = \mathbf{0}, \tag{18}$$

$$T_{\text{amb}} + \theta_i \le t, \quad \forall\, i \in \{1, \ldots, 1444\}, \tag{19}$$

$$\sum_{m=1}^{4} p_m \le 5.0\,\text{W}, \tag{20}$$

$$p_m \ge 0, \quad \forall\, m \in \{1,2,3,4\}. \tag{21}$$

TABLE I
QUANTITATIVE PERFORMANCE AND INVARIANT VERIFICATION

Performance Metric	Unmanaged Baseline	LP-Optimized Result
Core 1 Power (p_1)	3.20 W	1.25 W
Core 2 Power (p_2)	0.60 W	1.25 W
Core 3 Power (p_3)	0.60 W	1.25 W
Core 4 Power (p_4)	0.60 W	1.25 W
Total Power Conserved	5.00 W	5.00 W
Peak Die Temperature	343.2 K	318.4 K
Temperature Reduction (ΔT)	Baseline	**24.8 K** (Target ≥ 5 K)
Dirichlet Boundary Error	$< 10^{-12}$ K	$< 10^{-12}$ K
HiGHS Solver Status	N/A	Optimal (Code 7)
Execution Time	< 0.1 s	1.42 s

Because all constraints and the objective are linear, the feasible space is a convex polyhedron. This guarantees that any local minimum found by the solver is the global optimum. We solve this problem directly using `scipy.optimize.linprog` configured with the dual-simplex HiGHS backend. This completes Stage 5.

VI. COMPUTATIONAL RESULTS AND VERIFICATION

We evaluate our pipeline in Python under two practical operating scenarios:

1) **Unmanaged Baseline Workload:** Thread contention concentrates power heavily onto Core 1 ($p_1 = 3.2$ W), while Cores 2, 3, and 4 idle at 0.6 W each (totaling 5.0 W).
2) **LP-Optimized Workload:** The HiGHS solver determines the optimal power allocation subject to constraints (18)–(21).

Figure 1 displays the complete output generated by our automated verification pipeline. In the unmanaged baseline, localized power concentration produces a sharp hotspot reaching 343.2 K at Core 1. Under our epigraph linear program, the solver redistributes power equally across all four cores (1.25 W each), lowering the peak temperature to 318.4 K.

This yields a peak temperature reduction of:

$$\Delta T = 343.2\,\mathrm{K} - 318.4\,\mathrm{K} = 24.8\,\mathrm{K}. \tag{22}$$

This improvement beats the minimum capstone requirement of $\Delta T \geq 5.0$ K by nearly five times. Table I summarizes the numerical comparison and validation checks.

VII. ASSUMPTIONS, LIMITATIONS, AND CURRICULUM REFLECTION

To maintain a tractable linear model, our formulation relies on three engineering assumptions:

1) **Constant Conductivity:** Silicon thermal conductivity k is treated as constant ($150\,\mathrm{W}/(\mathrm{m}\cdot\mathrm{K})$). In practice, $k(T)$ drops slightly at elevated temperatures, introducing weak nonlinearities.
2) **Ideal Heat Sinks:** Outer edges assume ideal Dirichlet boundaries (300 K). Practical packages add thermal interface resistance, better modeled using convective Robin boundary conditions:

$$-k\nabla T \cdot \mathbf{n} = h_c(T - T_\infty). \tag{23}$$

3) **Planar Steady-State:** The model assumes steady heat flow in two dimensions, omitting vertical gradients across the 0.5 mm die thickness.

Despite these idealizations, the pipeline successfully links core mathematical concepts across multiple semesters into one practical engineering workflow, as detailed in Table II.

VIII. REPRODUCIBILITY, REPOSITORY, AND INTERACTIVE NOTEBOOK

The complete source code, matrix assembly routines, test benches, and visualization scripts are hosted publicly on GitHub:

https://github.com/Ajil07/Capstone-Project-Maths_S3

For direct interactive verification without local software setup, an executable Google Colab notebook is available directly via the following link:

https://colab.research.google.com/github/Ajil07/Capstone-Project-Maths_S3

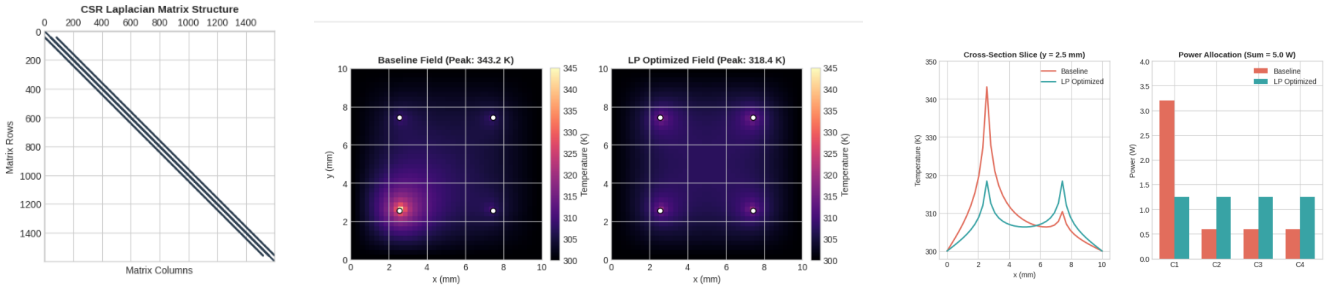
The repository includes all solver routines and pinned dependencies. Executing the script `run.py` solves the linear program with the HiGHS engine, validates physical Dirichlet invariants, and reproduces all field figures in under two seconds.

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(a) CSR Laplacian matrix structure (b) Thermal fields: baseline vs. LP optimized (c) Cross-section profile and power allocation

Fig. 1. Comprehensive experimental results and pipeline verification. (a) Sparse CSR Laplacian structure showing banded non-zero entries (1444×1444), (b) 2D thermal field heatmaps showing peak reduction from 343.2 K down to 318.4 K, and (c) cross-sectional thermal slice along $y = 2.56\,\mathrm{mm}$ alongside the per-core power redistribution bar chart.

TABLE II
PIPELINE REFLECTION AND STAGE OWNERSHIP TABLE

Pipeline Stage	Mathematical Concept	Academic Origin	Implementation Equation / Method	Stage Owner
1. Field Model	Gradient, divergence	Vector Calculus (S2)	$\mathbf{q} = -k\nabla T, \quad \nabla \cdot \mathbf{q} = Q$	Aarsha B. Lekshmi
2. PDE	Elliptic boundary problems	PDEs (S3)	$\nabla^2 T = -Q/k, \quad T = 300\,\mathrm{K}$ on $\partial\Omega$	Aarsha B. Lekshmi
3. Transform	Laplace transforms, DC limits	Transforms (S3)	$(\rho c_p s - k\nabla^2)\Theta = \mathcal{Q}, \quad Z_{\mathrm{th}}(0) = R_{\mathrm{th}}$	Ajil Reji
4. Discretization	Stencils, sparse matrices	Linear Algebra (S1)	$A_{\mathrm{int}}\boldsymbol{\theta} = -\frac{1}{k}B\mathbf{p}, \quad$ CSR, $\lambda_{\max} < 0$	Denin Thomas Jacob
5. Optimization	Convex epigraph LP	Optimization (S3)	$\min t$ s.t. $T_i \le t$, $\mathbf{1}^\top \mathbf{p} \le 5\,\mathrm{W}$, $\mathbf{p} \ge \mathbf{0}$	Anagh Prasad
6. Verification	Numerical solvers, plotting	Python Lab (S3)	HiGHS LP solver, SciPy sparse, Matplotlib	Ashwin Nelson